



What is Semi-Supervised Learning (SSL)?



- Mix of labeled and unlabeled data
- Useful when labeled data is scarce but unlabeled data is plenty
- Bridges gap between supervised (labeled) and unsupervised (no labels) learning



Why Expectation-Maximization (EM)?

- Helps train models using both labeled data
- for unlabeled, guess labels using on certain
- Iteratively improves the model to maximize likelihood of data



How EM Works (Step-by-Step)

- Initialize Model
- E-Step: For labeled data use known labels; for unlabeled, guess labels using model
- M-Step: Update model to fit both types
- Repeat until convergence



Evaluation

- Test final model on separate test data



Applications



Image classification (e.y. Google)

- NLP (e.g. Stanford NLP)
- Speech recognition



Challenges

- Assumes missing data is random
- Risk of local optima
- Slow for large datasets

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What is K-Nearest Neighbors (KNN)?



A simple algorithm that classifies data based on how close it is to others

Example: To guess if a new user will cancel a subscription, KNN checks how similar users behaved

Why is KNN called a Lazy Learner?

No training phase — it just stores data
Makes predictions only when needed
by checking nearby data points

How Does KNN Work? (Steps)

- Choose K: "K" = number of neighbors

- Measure Distance:

- Euclidean (straight line)
- Mankattan (grid-based)
- Minkowski (generalized)

- Find Nearest Neighbors: sort points by distance

- Predict: majority vote for classification or average for regression

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How to Choose the Best K?



→ Too small → overfitting

→ Too big → underfitting



Use odd values, cross-validation, elbow method



KNN for Classification vs Regression



Classification	Regression
Majority voting	→ Mean value
Class label	→ Numeric output



Applications

→ Recommendation Systems

– Netflix, Amazon

🕒 Spam Detection

👤 Customer Segmentation

🗣️ Speech Recognition

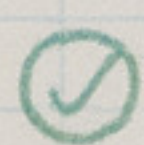
⚠️ Disadvantages



– Slow for large datasets

• Struggles with high-dimensional data

• Sensitive to noise and irrelevant features



Advantages

- Easy to understand and implement
- No training required
- Works for both classification & regression



KNN vs Other Algorithms

Decision Trees	Better
Random Forests	Less
SVM	Powerful
Neural Nets	Large



Final Summary

- Easy to understand and implement
- No training required
- Works for both classification & regression



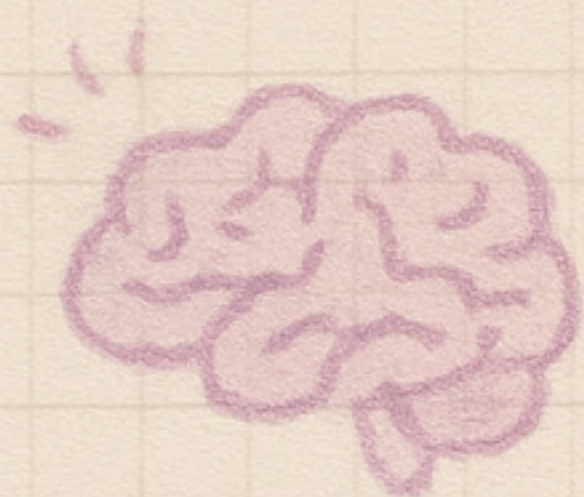
Final Summary



KNN = classify by training data
 Great for small datasets and quick testing
 Choose k wisely
 Good for recommendation and filtering



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What are Artificial Neural Networks (ANNs)?

- Inspired by the human brain: **billions of neurons**
- ANNs learn from data, not hand-coded rules

Structure of ANNs

Brain Part	ANN Equivalent
Dendrites	Inputs
Soma (cell body)	Node/Neuron
Synapses	Weights
Axon	Output
Synaptic plasticity	Backpropagation



Layers of ANN:

- Input Layer - receives
- Hidden Layers - process
- Output Layer - gives result

How ANN Learns (Training Process)

- Input Training Data (eg, cat images)
- Pattern Recognition (features like whiskers, ears)
- Testing & Validation

How ANN Learns (Training Process)



Types of Neural Networks

- Feedforward Neural Network
Input → Hidden → Output
- Convolutional Neural Network (CNN) - image-focused,
- Recurrent Neural Network (RNN) (for memory)

Recurrent Neural Network (RNN) → Pooling

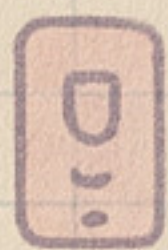
Modular Neural Network → Network AIN

Radial Basis Function Network (RBFN) →

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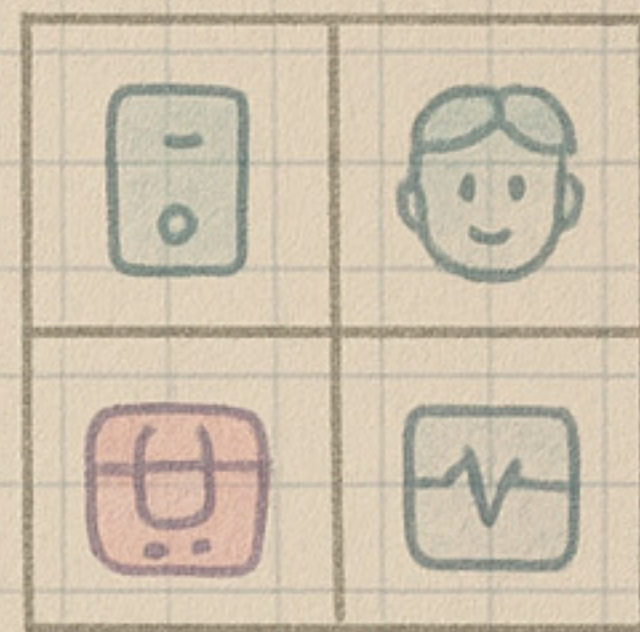


Applications of ANNs



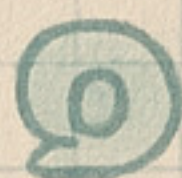
Social Media:

- friend suggestions, recommendations, facial recognition



Marketing & Sales:

- personalized ads



Healthcare

- cancer detection
patient monitoring



Personal Vision:

voice recognition
NLP, generating answers



Personal Assistants

- voice recognition
NLP, generating answers



NLP = translation, sentiment

chatbots



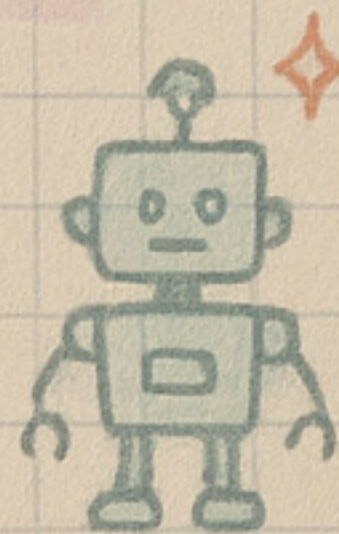
Computer Vision

- object detection
face recognition



The Future of Neural Networks

- Faster models (100x better)
- Giant models (108+ parameters)
- Huge economic value (\$15.7 trillion by 2030)
- Improved transparency + hybrid AI



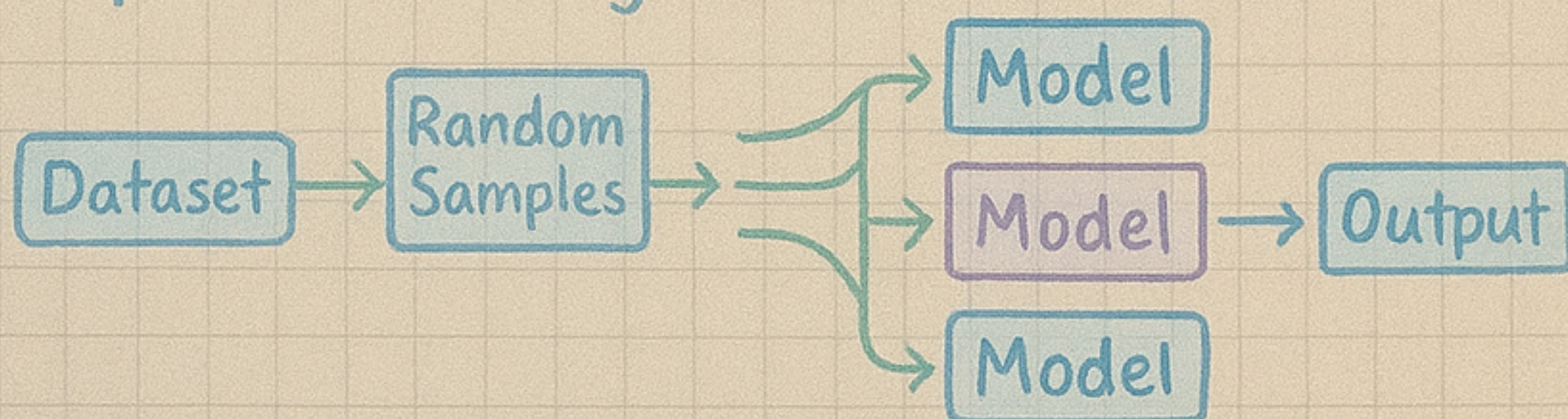
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🌸 What is Bagging?

Bagging = Bootstrap Aggregating
Ensemble method

- Trains many models on random data subsets (sampled with replacement)
- Combines their outputs using voting (classification) or averaging (regression)

Goal: reduce variance, prevent overfitting, improve stability

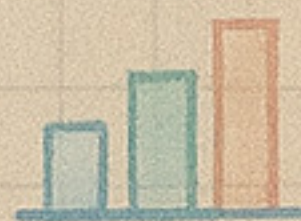


🕒 Steps in Bagging

- Prepare dataset
- Create bootstrapped samples
- Train base models on each
- Aggregate: majority vote / average
- Evaluate performance
- Tune with cross-validation

🧩 Ensemble Methods Overview

- Bagging: parallel training
- Boosting: sequential fixing
- Random Forest: bagging + trees + random features



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Why Bagging Works

- Reduces variance
- Prevents overfitting
- Improves generalization



Doodle idea: A stormy model calmed by ensemble voting icons

Applications

- **Classification:** Medical diagnosis, churn prediction, spam detection
- **Regression:** Housing prices, stock forecasting
- **Anomaly Detection:** Fraud, network security
- **Feature Selection:** Highlights key-features
- **Imbalanced Data:** Handles class imbalance
- **Time-Series:** Forecasting, segmentation



Python Example (Sklearn)

- Use BaggingClassifier + DecisionTreeClassifier on Iris
- Evaluate using accuracy_score

✓ Conclusion:

Bagging = stable, accurate, and generalizable

- ✓ Great for noisy,
- ✓ complex data

Feature	Bagging	Boosting
Training Type	Parallel	Sequential
Focus	Variance	Bias
Sensitivity	Low	High
Speed	Faster	Slower
Examples	Random Forest	XGBoost, AdaBoost

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What Are Recommender Systems?

- Predict what user might like next
- Based on past behavior: clicks, purchases, views

🔦 Purpose:

- Help users choose in option-heavy environments
- Ex: Netflix → shows, Amazon → products, Spotify: playlists

🌀 Types:

- Collaborative Filtering (user-based & item-based)
- Content-Based Filtering (uses item features)
- Hybrid (combines both)

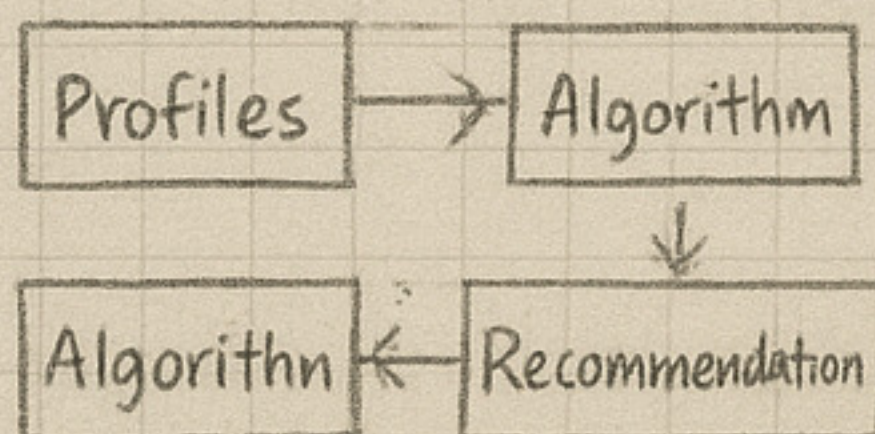
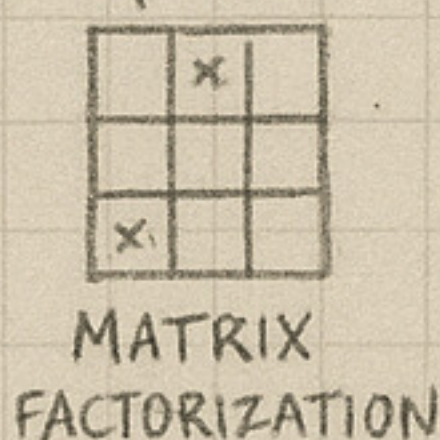
👍 Benefits:

- Speed decisions (ratings)
- Increase engagement
- Improves satisfaction
- Boosts retention

Business Impact	
Metric	Impact
Revenue	+35% suggestions
Netflix Views	+6% via rec system
CTR	+60% w/ personalization

⚠️ Advanced Techniques

- Cold start
- Filter bubble
- Data sparsity
- Scalability
- Privacy



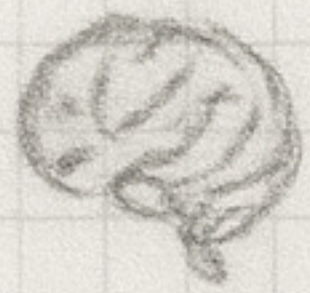
🌀 Future

- Context-aware (mood, time)
- Explainable AI
- Federated learning
- Multimodal input
- (text; image, audio)

— Power combo of math + ML + behavior
Drives Netflix, Amazon, YouTube
Moving toward smarter, safer systems

Drives Netflix, Amazon, YouTube.
Moving toward smarter, safer systems

Drives Netflix, Amazon, YouTube



What is Reinforcement Learning?

- Agent interacts with environment
- Learns from feedback: rewards & penalties
- Goal: maximize cumulative reward over time



RL vs Supervised Learning:

- No labeled data
- Learns through trial & error
- Exploration vs. Exploitation

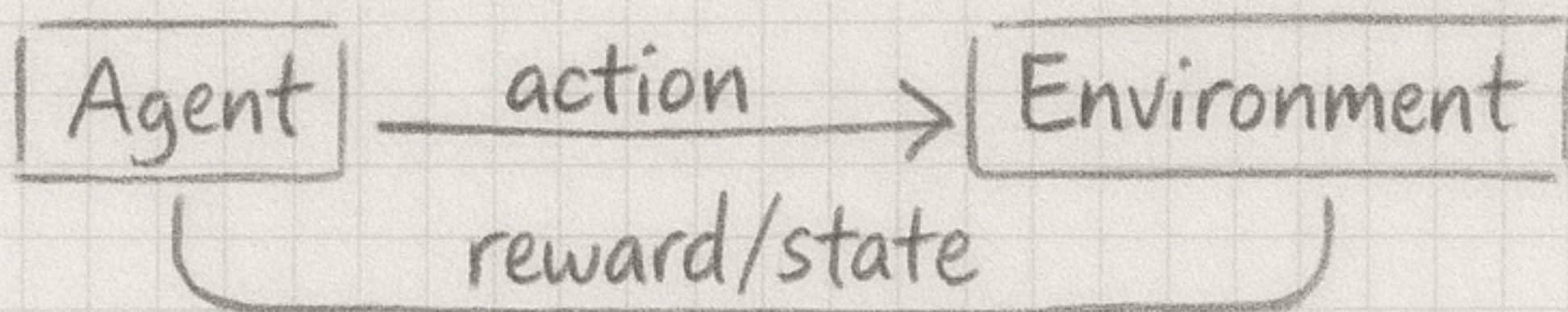


Key Building Blocks:

Term	Example
Agent	Robot in warehouse
Environment	Warehouse layout
State	Robot's position
Action	Move, turn, pick item
Reward	+10 delivery, -5 wall hit



RL Process (Diagram):



Key Mathematical Concepts:

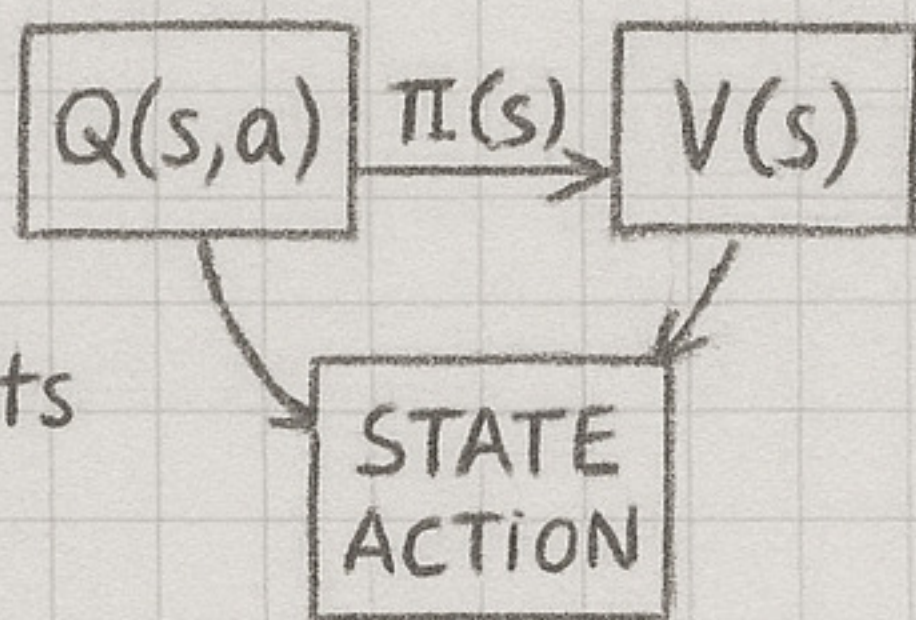
- Policy $\pi(s)$: action to take in state s
- Reward Function $R(s, a)$: value of action in a
- Value Function $V(s)$: future total reward
- Q-Value $Q(s, a)$: reward expected from action a in state s

Important Algorithms:

- Q-Learning $\rightarrow$ off-policy, learns best $Q(s, a)$
- DQN $\rightarrow$ Q-learning + neural nets
- Policy Gradient $\rightarrow$ directly optimize policy
- Actor-Critic $\rightarrow$ uses both policy & value

Applications of RL:

-  Robotics: walk, grasp
-  Gaming: AlphaGo, Dota bots
-  Autonomous Driving
-  Finance: trading bots
-  Healthcare: treatments
-  Smart Grid
-  Recommendations (YouTube, Netflix)



 Challenges: Exploration vs exploitation
Delayed rewards

 Summary: Deep environments = more compute

RL = agent learning from interaction

Rewards = feedback

Powerful for real-world, complex decision problems

Exploration vs. Exploitation

- Exploitation: choose known high-reward actions (e.g, repeat best chess move)
- Exploration: try unknown actions to discover better ones

Goal: balance for long-term reward, avoid local optima or wasted effort

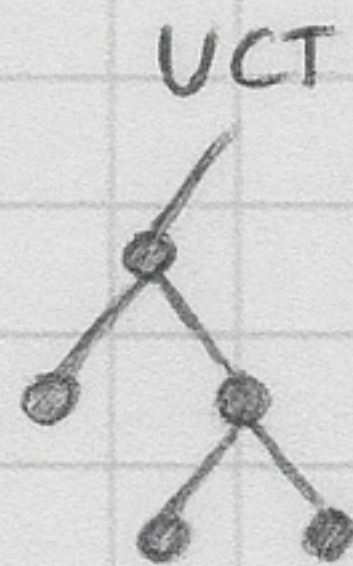
Monte Carlo Inference:

- Monte Carlo Methods
- Reinforcement Learning
- Bandit Problems
- Game AI, Robotics



Monte Carlo Tree Search (MCTS)

- Selection - follow best UCT path
- Expansion - add child nodes
- Simulation - rollout randomly
- Backpropagation - update rewards



UCT Score Formula:

$$\text{Score} = \bar{X}_j + c \cdot \sqrt{\frac{\ln N}{n_j}}$$

$\bar{X}_j$: avg reward (exploitation)

$c \cdot \sqrt{\dots}$: exploration bonus

Algorithms:

- UCT (used in MCTS)
- Epsilon-Greedy: ϵ chance of random move
- Annealing: reduce unknowns over time

Game AI & Planning:

AlphaGo: MCTS + Neural Nets

Too much exploitation
= predictable

Too much exploration
= weak

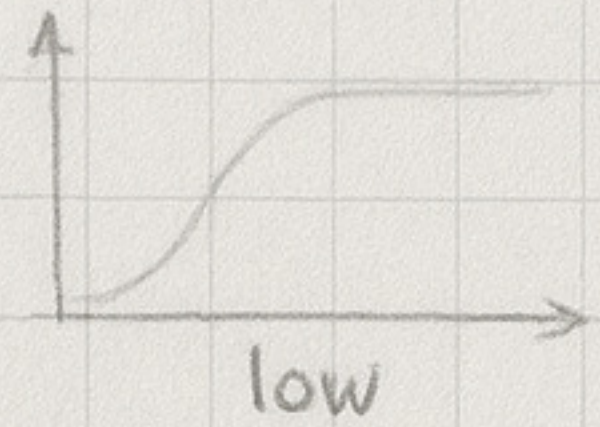
Planning: explore unknowns
vs follow safe paths

🔥 Probabilistic Methods:

- Softmax / Boltzmann: randomize action choice by rewards
High temp $\rightarrow$ more randomness
- Thompson Sampling: sample based on uncertainty
- Bayesian Optimization: pick based on value & uncertainty
- Contextual Bandits: explore based situation

📅 Multi-Armed Bandits:

- Pick between unknown options
Goal: maximize cumulative reward
- Used in A/B testing, ads, trials



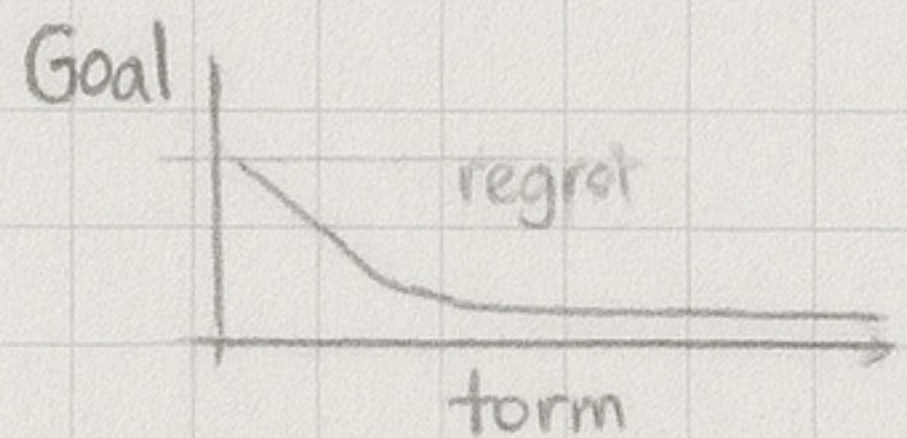
⚙️ Tuning Table:

- Convergence: all strategies
 $\rightarrow$ correct if explored enough
- Regret bounds
 $\rightarrow$ many algorithms grow regret slowly (e.g., $\log T$)

Tuning Table	
C	UCT exploration
E	Random choice rate
T	Softmax randomness
Prior	Initial belief weight domain-specific

🔗 Emerging Strategies:

- Meta-learning
- Info gain (entropy)
- Intrinsic curiosity
- Hierarchical exploration

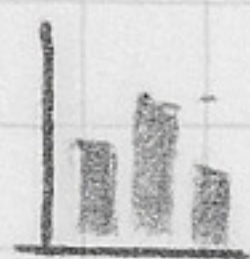


📝 Final Takeaways:

- Balance is key
- Algorithms: UCT, ϵ -greedy, Thompson, Softmax
- Used in AlphaGo, robots, ads, RL

📝 Final Takeaways:

- Balance is key
- Algorithms: UCT, ϵ -greedy, Thompson, Softmax
- Future = adaptive, efficient, safe exploration



What is a Cost Function?

- Measures how wrong the model is
- Ex: distance between prediction + actual

Goal: Minimize cost $\rightarrow$ better model

Most Common: Mean Squared Error (MSE)

$$J(\theta) = \frac{1}{2m} \sum (h_{\theta}(x) - y)^2$$

$h_{\theta}(x)$ = prediction, y = actual

m = # of examples



What is Gradient Descent?

Finds minimum of cost function

Updates parameters θ using slope

Formula:

$$\theta_j := \theta_j - \frac{\partial J(\theta)}{\partial \theta_j}$$

α = learning rate

∂J = gradient



Learning Rate Table:

α Value
Slow convergence

α Value	Result
Too Small	Slow convergence
Too Big	Overshoots, unstable
Just Right	Fast, stable learning



Variants of Gradient Descent:

Batch GD: all data per step $\rightarrow$ stable, slow

SGD: 1 example per step $\rightarrow$ fast, noisy

Mini-Batch: small groups $\rightarrow$ balanced

Example: Learn $y = 2x$

$$x = [1, 2, 3], y = [2, 4, 6]$$

$$\text{Init: } \theta_0 = 0, \theta_1 = 0, \alpha = 0.1$$

→ After 1 iter. → $\theta_0 \approx 0.4, 9333$

$$\rightarrow \text{Line: } h_0(x) \approx 0.4 + 0.933x$$

⌘ Gradient Descent Loop:

Predict → $h_0(x)$

Compute cost → $J(\theta)$

Compute gradients → $\partial J / \partial \theta$

Update params → $\theta_j: \theta_j := \theta_j - \alpha \cdot \nabla$

⌘ Advanced Optimizers:

Momentum

AdaGrad per-parameter learning

RMSProp better

Adam → adaptive

Cost Functions by Problem

Regression

MSE

Binary Classify

Binary Cross-entropy

Multi-Class

Categorical Cross-Entropy

⌘ Tips:

Normalize features

Good θ init

Set stopping rules

Monitor cost graph

Python Demo:

After 100 iters: $\theta_0 \approx 0, \theta_1 = 2$

→ close to $y = 2x$

Final Takeaways:

Cost → how wrong

GD → corrects step by step

Regularize to generalize

⌘ Regularization

Penalty for complexity

$$J_{\text{reg}}(\theta) = J(\theta) + \lambda \cdot R(\theta)$$

L_1 : sum of θ

L_2 : sum of θ^2

Elastic Net = $L_1 + L_2$

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